- $\mathrm{Var}(X) = E\left[X^2\right] - E[X]^2$.
- $\mathrm{Cov}(X,Y) \coloneqq \mu_{XY} - \mu_X\mu_Y = E\big[(X-\mu_X)(Y-\mu_Y)\big]$
- **Markov**: if $X \geq 0$, then $P(X \geq a) \leq \frac{E[X]}{a} \forall a > 0$.
- **Chebyshev**: if $\sigma^2 < \infty$, $P\big(g(X) > a^2\sigma^2\big) \leq \frac{E[g(X)]}{a^2\sigma^2}$.
- Total probability: if $\{B_i : i \in I\}$ is a partition of the sample space, then
$$P(A) = \sum_{i\in I} P(A \mid B_i)\, P(B_i).$$
- Total expectation: $E[X] = E\big[E[X \mid Y]\big]$.
- Total var: $\mathrm{Var}(X) = E[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(E[X \mid Y])$.
- Bayes's Theorem: $P(A \mid B) = \frac{P(B|A)P(A)}{P(B)}$.
- Markov chain: future independent of the past, i.e., X_{n+1} is at most dependent on X_n.
- Transition probability: $p_{ij}^{n,m} = P(X_m = j \mid X_n = i)$.
- Transition probability matrix: if π_i is the distribution of X_i, then $\pi_t = \pi_0 \prod_{i=0}^{t-1} \boldsymbol{P}^{i,i+1}$.
- Stationary Markov chain: TPM is independent of the timestamp n.
- **Stochastic matrix**: non-negative matrix such that row sums are 1.
- **Chapman-Kolmogorov**: $\boldsymbol{P}^{(m)} = \boldsymbol{P}\boldsymbol{P}^{(m-1)} = \boldsymbol{P}^{(m-1)}\boldsymbol{P}$. If X is stationary, then for all $m, n \in \mathbb{N}$,
$$P_{ij}^{0,m+n} = \sum_{k\in S} P_{ik}^{0,m} P_{kj}^{0,n},$$
and $P(X_n = j \mid X_0 = i) = (\boldsymbol{P}^n)_{ij}$.
- if $\mu \coloneqq \begin{bmatrix} f(x_1) & f(x_2) & \cdots & f(x_s)\end{bmatrix}^{\mathrm{T}}$ for some function f on the state space, then
$$(\boldsymbol{P}^n\mu)_i = \sum_{j=1}^{s} \boldsymbol{P}_{ij}^n \mu_j = \sum_{j=1}^{s} P(X_n = x_j \mid X_0 = x_i)\, f(x_j) = E[f(X_n) \mid X_0 = x_i].$$
Suppose $X_0 \sim \lambda$, then clearly
$$E[f(X_n)] = \sum_{i=1}^{s} E[f(X_n) \mid X_0 = x_i]\,\lambda_i = \sum_{i=1}^{s} \lambda_i (\boldsymbol{P}^n\mu)_i = \lambda \boldsymbol{P}^n \mu.$$
- $\boldsymbol{P}\boldsymbol{1} = \boldsymbol{1}$ so 1 is an eigenvalue of $\boldsymbol{P}^{\mathrm{T}} \implies \exists$ stationary distr.
- **Stationary distribution**: $\pi = \pi\boldsymbol{P}$.
- If λ is an eigenvalue of $\boldsymbol{P}$, then $|\lambda| \leq 1$.
- **Absorbing state**: for all $j \neq i$, we have $P_{ij} = 0$.
- $x \leftrightarrow y \iff \exists m, n \in \mathbb{N}$ such that $P_{xy}^{(m)}, P_{yx}^{(n)} > 0$.
- **Irreducible chain**: only one class.
- **Return prob.**: $P_{ii}^{(n)} = P(X_n = i \mid X_0 = i)$.
- **1st return prob.**: $f_{ii}^{(n)} = P(X_j \neq i \forall j < n, X_n = i \mid X_0 = i)$. $f_{ii}^{(0)} = 0$, $f_{ii}^{(n)} \leq P_{ii}^{(n)}$.
- $P_{ii}^{(n)} = \sum_{k=0}^{n} f_{ii}^{(k)} P_{ii}^{(n-k)}$.
- $f_{ii} = \sum_{n=0}^{\infty} f_{ii}^{(n)}$ is the prob. of returning to i in finite time. i is **recurrent** if $f_{ii} = 1$ and **transient** if $f_{ii} < 1$.
- Recurrent: $P\left(\sum_{n=1}^{\infty} I\{X_n = i\} = \infty \mid X_0 = i\right) = 1$ so
$$E\left[\sum_{n=0}^{\infty} I\{X_n = x\} \mid X_0 = x\right] = \sum_{n=0}^{\infty} \boldsymbol{P}^n(x,x) = \infty,$$
but $\boldsymbol{P}^n(x,x)$ may converge to 0.
- Number of revisits to i: $N_i \sim \mathrm{Geo}(1 - f_{ii})$. $E[N_i \mid X_0 = i] = \frac{f_{ii}}{1-f_{ii}}$ and expected number of visits including the initial one is $\frac{1}{1-f_{ii}}$.
- i is a transient state iff $\sum_{n=1}^{\infty} P_{ii}^{(n)}$ is finite.
- If i is transient, then $\lim_{m\to\infty} \sum_{n=m}^{\infty} P_{ii}^{(n)} = 0$ by monotone convergence theorem.
- Irreducible recurrent: $P\left(\sum_{n=0}^{\infty} I\{X_n = i\} = \infty\right) = 1$
- Irreducible transient: $P\left(\sum_{n=0}^{\infty} I\{X_n = i\} < \infty\right) = 1$
- All finite-state irreducible chains are recurrent.
- Reducible chains will enter one of the recurrent classes in the long-run.
- **Period**: $d(i) \coloneqq \gcd\left\{n \in \mathbb{N}^+ : P_{ii}^{(n)} > 0\right\}$. i is **aperiodic** iff $d(i) = 1$.
- If $i \leftrightarrow j$, then $d(i) = d(j)$.
- $\forall i \in S, \exists N \in \mathbb{N}$ s.t. $\forall n \geq N$, $P_{ii}^{(n\cdot d(i))} > 0$ and $P_{ji}^{(m+n\cdot d(i))} > 0$ whenever $P_{ji}^{(m)} > 0$.
- If $\boldsymbol{P}$ is the TPM for a finite-state irreducible aperiodic chain, then $\exists N \in \mathbb{N}^+$ such that $\boldsymbol{P}^{(N)}$ has all positive entries (definition for regular chain).
- Irreducible, aperiodic, finite-state $\implies$ regular. Regular $\implies$ irreducible.
- $\boldsymbol{P}^{(k)}$ is regular $\implies \boldsymbol{P}^{(n)}$ is regular $\forall n \geq k$.
- Let $\boldsymbol{P}$ be a regular TPM for some regular Markov chain with finite state space, then there is a unique stationary distribution π with $\pi_j = \lim_{n\to\infty} P_{ij}^{(n)}$ for all i, j. $\pi_j = P(X_n = j)$ is the proportion of time where $X_n = j$ in the long-run.
- Local bal.: $\pi_i P_{ij} = \pi_j P_{ji}$ for all $i, j \implies$ global bal..
- **Stopping time**: $T_A \coloneqq \min\{n \in \mathbb{N} : X_n \in A\}$ is the first time X enters A.
- **Null recurrent**: $\mathbb{E}[T_i \mid X_0 = i] = \infty$. **Positive recurrent**: $\mathbb{E}[T_i \mid X_0 = i] < \infty$.
- **Ergodic**: irreducible aperiodic positive recurrent.
- Define $m_i \coloneqq \mathbb{E}[R_i \mid X_0 = i] = \sum_{n=1}^{\infty} n f_{ii}^{(n)}$ in ergodic chains, then $\lim_{n\to\infty} P_{ij}^{(n)} = \lim_{n\to\infty} P_{jj}^{(n)} = \frac{1}{m_j} = \pi_j$ gives a stationary distribution.

- Null recurrent: $\lim_{n\to\infty} P_{ij}^{(n)} = 0$.
- Irreducible recur.: $\lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} P_{ij}^{(k)} = \frac{1}{m_j}$. If aperiodic: $\lim_{n\to\infty} P_{ij}^{(n)} = \frac{1}{m_j}$; else $\lim_{n\to\infty} P_{jj}^{(nd)} = \frac{d}{m_j}$.
- If $f(x) = P(T_A < T_B \mid X_0 = x)$, then for all $x \notin A \cup B$,

$$\begin{aligned} f(x) &= \sum_{y\in S} P(T_A < T_B \mid X_1 = y, X_0 = x) P(X_1 = y \mid X_0 = x) \\ &= \sum_{y\in S} P(T_A < T_B \mid X_0 = y) P(X_1 = y \mid X_0 = x) \\ &= \sum_{y\in S} P_{xy} f(y). \end{aligned}$$

- **First-step analysis**:
  1. Identify quantity of interest $a_i(T) = h(i, X_1, \cdots, X_T \mid X_0 = i)$.
  2. $a_i(T) = \sum_{k\in S} h(\cdot \mid X_1 = k, X_0 = i) P(X_1 = k \mid X_0 = i)$.
  3. Consider $Y_n = X_{n+1}$ and establish $h(\cdot \mid X_1 = k, X_0 = i) = g_i\big(a_k(T)\big)$.
  4. Solve the system.
- Gambler's ruin: $X_0 = k$ for $0 < k < N$ with winning probability p.
  - Fair game: $P(X_T = 0 \mid X_0 = k) = 1 - \frac{k}{N}$ and $E[T \mid X_0 = k] = k(N-k)$.
  - Otherwise:

$$P(X_T = 0 \mid X_0 = k) = 1 - \frac{1-(q/p)^k}{1-(q/p)^N},$$

$$E[T \mid X_0 = k] = \frac{1}{p-q}\left[\frac{N\left(1-(q/p)^k\right)}{1-(q/p)^N} - k\right]$$

- Random walk: $\frac{\xi_i+1}{2} \sim \text{Bernoulli}(p)$, $\frac{X_n+n}{2} \mid X_0 = 0 \sim \text{Bin}(n,p)$.
- **Page rank**: by stationary distr., $\pi_{n+1} = \lambda\pi_n \boldsymbol{P} + (1-\lambda)\pi_n$ if perturbation is needed.
- Hastings-Metropolis:
  - p is target distribution with sample space S and M is the max depth of iterations.
  - Fix an irreducible chain with state space S and TPM $\boldsymbol{Q}$ which is symmetric. Select some $X_0 \in S$.
  - At the n-th iteration:
    * Generate $Y \sim q$ s.t. $q_j = Q_{X_n j}$ for all $j \in S$ and $U \sim \text{U}(0,1)$.
    * Compute $\alpha(X_n, Y) := \min\left\{\frac{p_Y Q_{Y X_n}}{p_{X_n} Q_{X_n Y}}, 1\right\}$.
    * $U < \alpha(X_n, Y)$? $X_{n+1} = Y$: $X_{n+1} = X_n$.
  - Return $Z := X_M \sim p$ approximately.
- $P(Z_{n+1} = j \mid Z_n = i) = Q_{ij}\alpha(i,j) = P_{ij}$
- $P(Z_{n+1} = i \mid Z_n = i) = Q_{ii} + \sum_{k\neq i} Q_{ik}\big(1-\alpha_{ik}\big) = P_{ii}$
- If $p = c\boldsymbol{b}$ where $\boldsymbol{b}$ is a kernel function, only need $\frac{b_i}{b_j}$.
- Prob. gen. func. $\phi_X(t) := \mathbb{E}\left[t^X\right] = \sum_k P(X=k)t^k$. $\mathbb{E}[X] = \phi'(1)$ and $\mathbb{E}\left[X^2\right] = \phi''(1) + \phi'(1)$.
- Random sum: $\text{Var}(S) = \mu_N \text{Var}(X) + \sigma_N^2 \mathbb{E}[X]$.
- $\phi_X^{(n)}(t) = \sum_{k=n}^{\infty} k! P(X=k) t^{k-n}$, so $P(X=x) = \frac{\phi_X^{(x)}(0)}{x!}$
- $\phi_{X+Y} = \phi_X \phi_Y$ if independent.
- $\phi_{X_{n+1}\mid X_n = x}(t) = \mathbb{E}\left[t^{X_{n+1}} \,\middle|\, X_n = x\right] = \mathbb{E}\left[\prod_{i=1}^{x} t^{\xi_i}\right] = \prod_{i=1}^{x}\phi_{\xi_i}(t)$
- $\phi_{X_n\mid X_0=k}(t) = \phi_{X_1\mid X_0=k}\big(\phi_\xi^{n-1}(t)\big)$, so $\phi_{X_n\mid X_0}(t) = \phi_\xi^n(t)^{X_0}$
- $\mathbb{E}[X_n \mid X_0] = \mu^n X_0$ and $\text{Var}(X_n \mid X_0) = \begin{cases} \frac{1-\mu^n}{1-\mu}\left(\mu^{n-1}\sigma^2\right) X_0 & \text{if } \mu \neq 1 \\ n\left(\mu^{n-1}\sigma^2\right) X_0 & \text{if } \mu = 1 \end{cases}$.
- **Extinction Prob.**: $u_n^{(k)} := P(X_n = 0 \mid X_0 = k) = u_n^k$
- $u_n = \sum_{i\in S} P(X_1 = i \mid X_0 = 1) u_{n-1}^{(i)} = \sum_{i\in S} P(\xi = i) u_{n-1}^i$
- $u_n^{(k)} = \phi_\xi^{n-1}\big(P(\xi=0)\big)^k$
  - if $P(\xi = 0) = 0$, then $u_\infty^{(k)} = 1$;
  - if $P(\xi = 0) > 0$, then if $\mathbb{E}[\xi] \leq 1$, $u_\infty^{(k)} = 1$; else, $u_\infty \in [0,1)$ is s.t. $u_\infty = \phi_\xi(u_\infty)$
- Pois. Proc.: $X(0) = 0$, $X(s+t) - X(s) \sim \text{Pois}(\lambda t)$ and for any $0 = t_0 < t_1 < \cdots < t_n$, $X(t_i) - X(t_{i-1})$'s are mutually independent
- Pois. Pt. Proc.:
  - for any $0 = t_0 < t_1 < \cdots < t_n$, $N\big((t_i, t_{i+1}]\big)$'s are mutually independent;
  - $P\Big(N\big((t,t+h]\big) \geq 1\Big) = \lambda h + o(h)$ and $P\Big(N\big((t,t+h]\big) \geq 2\Big) = o(h)$ as $h \to 0$.
- Waiting time: $W_n \sim \text{Gamma}(n, \lambda)$ s.t. $f_{W_n}(t) = \frac{\lambda^n t^{n-1}}{(n-1)!}\mathrm{e}^{-\lambda t}$.
- Sojourn time: $S_n = W_1 \sim \text{Exp}(\lambda)$ where $f_{S_n}(t) = \lambda \mathrm{e}^{-\lambda t}$
- $W_1 \mid X(t) = 1 \sim \text{U}(0,t)$. Joint is the sum of independent uniform r.v.

$$\begin{aligned} f\big(w_1, w_2, \cdots, w_n \mid X(t) = n\big) &= \frac{P(S_n > t - w_n)\prod_{i=1}^{n} f_{S_{i-1}}(w_i - w_{i-1})}{\frac{(\lambda t)^n \mathrm{e}^{-\lambda t}}{n!}} \\ &= \frac{n!\mathrm{e}^{-\lambda(t-w_n)}\prod_{i=1}^{n}\lambda \mathrm{e}^{-\lambda(w_i - w_{i-1})}}{(\lambda t)^n \mathrm{e}^{-\lambda t}} = \frac{n!}{t^n}. \end{aligned}$$

- $f_{W_k\mid X(t)=n}(x) = \frac{n! x^{k-1}(t-x)^{n-k}}{(n-k)!(k-1)! t^n}$